

Bulan 6: Deployment & DevOps

Minggu 1: Production Settings & Preparation

Hari 1: Production Settings Configuration

Topik: DEBUG=False, ALLOWED_HOSTS, security settings

Proyek: ProductionSettings

Configure production settings

- Separate settings files (base, dev, prod)
- DEBUG configuration
- ALLOWED_HOSTS setup
- SECRET_KEY dari environment
- Security middleware
- SECURE_SSL_REDIRECT
- Teknologi: Django production settings

Hari 2: Static Files in Production

Topik: collectstatic, STATIC_ROOT, WhiteNoise

Proyek: StaticFilesProduction

Serve static files in production

- Install WhiteNoise
- Configure STATIC_ROOT
- collectstatic command
- Compression & caching
- CDN simulation
- Teknologi: WhiteNoise, static files

Hari 3: Database Configuration for Production

Topik: PostgreSQL setup, connection pooling, database backup

Proyek: PostgreSQLSetup

Production database setup

- Install psycopg2

- PostgreSQL configuration
- Connection pooling dengan psycopg2
- Database migrations
- pg_dump basics (backup simulation)
- Teknologi: PostgreSQL, database config

Hari 4: Gunicorn & WSGI

Topik: WSGI, Gunicorn configuration, workers

Proyek: GunicornSetup

Production WSGI server

- Install Gunicorn
- gunicorn.conf.py
- Configure workers
- Bind to socket
- Timeout settings
- Worker class
- Teknologi: Gunicorn, WSGI

Hari 5: Nginx Basics

Topik: Reverse proxy, Nginx configuration, upstream

Proyek: NginxConfig

Nginx as reverse proxy

- Install Nginx (simulation)
- nginx.conf basics
- Proxy pass to Gunicorn
- Static files serving
- Media files serving
- Gzip compression
- Teknologi: Nginx configuration

Hari 6: SSL/TLS & HTTPS

Topik: Let's Encrypt, SSL certificates, HTTPS configuration

Proyek: SSLSetup

Setup HTTPS

- SSL certificate (self-signed untuk testing)
- Nginx SSL configuration
- Force HTTPS redirect
- HSTS headers
- SSL best practices
- Teknologi: SSL/TLS, HTTPS

Hari 7: Review & Mini Project

Mini Project: ProductionReadyBlog

Blog application production-ready

Features:

- Separate settings (dev, staging, prod)
- PostgreSQL database
- Static files dengan WhiteNoise
- Gunicorn as application server
- Nginx configuration
- SSL setup (self-signed)
- Environment variables
- Production logging
- Security headers
- Error pages (404, 500)
- Health check endpoint
- Teknologi: PostgreSQL, Gunicorn, Nginx, SSL, Production settings

Minggu 2: Containerization & Docker

Hari 8: Docker Basics

Topik: Docker concepts, images, containers, Dockerfile

Proyek: DockerIntro

Introduction to Docker

- Install Docker
- Basic Docker commands (run, ps, stop)
- Pull images
- Run containers
- Container lifecycle
- Docker hub basics
- Teknologi: Docker fundamentals

Hari 9: Dockerfile for Django

Topik: Writing Dockerfile, base images, layers, best practices

Proyek: DjangoDockerfile

Containerize Django app

- Create Dockerfile
- Choose base image (python:3.11-slim)
- Copy requirements.txt
- Install dependencies
- COPY application code
- EXPOSE port
- CMD dengan Gunicorn
- Teknologi: Dockerfile, Django container

Hari 10: Docker Compose

Topik: docker-compose.yml, services, networks, volumes

Proyek: DjangoCompose

Multi-container setup

- docker-compose.yml
- Web service (Django)
- Database service (PostgreSQL)
- Redis service
- Environment variables
- Volumes for persistence
- Networks

- Teknologi: Docker Compose

Hari 11: Docker Volumes & Data Persistence

Topik: Named volumes, bind mounts, data management

Proyek: DataPersistence

Manage data in Docker

- Named volumes untuk database
- Bind mounts untuk development
- Volume backup
- Data migration between volumes
- Clean up unused volumes
- Teknologi: Docker volumes

Hari 12: Docker Networking

Topik: Bridge networks, custom networks, service discovery

Proyek: DockerNetworking

Container networking

- Create custom networks
- Connect services
- Service discovery by name
- Network isolation
- Port mapping
- Teknologi: Docker networking

Hari 13: Multi-stage Builds

Topik: Multi-stage Dockerfile, optimization, smaller images

Proyek: OptimizedDockerfile

Optimize Docker images

- Multi-stage build
- Build stage vs runtime stage
- Reduce image size
- .dockerignore file

- Layer caching
- Teknologi: Docker optimization

Hari 14: Review & Mini Project

Mini Project: MicroservicesSetup

Microservices dengan Docker Compose

Services:

- API service (Django DRF)
- Database service (PostgreSQL)
- Cache service (Redis)
- Worker service (Celery)
- Nginx service (reverse proxy)

Features:

- docker-compose.yml dengan 5 services
- Custom networks
- Named volumes untuk data
- Environment variables per service
- Health checks
- Service dependencies
- Development & production compose files
- Teknologi: Docker Compose, Multi-container, Microservices pattern

Minggu 3: CI/CD & Automation

Hari 15: Git & Version Control Review

Topik: Git workflow, branching strategies, gitignore

Proyek: GitWorkflow

Production Git workflow

- Feature branch workflow
- main, develop, feature branches
- Pull requests simulation
- .gitignore for Django

- Commit message conventions
- Teknologi: Git best practices

Hari 16: GitHub Actions Basics

Topik: Workflows, jobs, steps, triggers

Proyek: GitHubActionsIntro

CI/CD dengan GitHub Actions

- Create .github/workflows/ci.yml
- Trigger on push & PR
- Checkout code
- Setup Python
- Install dependencies
- Basic workflow
- Teknologi: GitHub Actions

Hari 17: Automated Testing in CI

Topik: Run tests in CI, test coverage, pytest

Proyek: TestingPipeline

Automated testing workflow

- Run Django tests in CI
- pytest configuration
- Coverage report
- Fail on test failures
- Test multiple Python versions
- Teknologi: CI testing, pytest

Hari 18: Linting & Code Quality

Topik: flake8, black, isort, pre-commit hooks

Proyek: CodeQuality

Code quality automation

- Install flake8, black, isort

- Configure each tool
- CI workflow untuk linting
- Pre-commit hooks
- Fail CI on linting errors
- Teknologi: Code quality tools

Hari 19: Build & Push Docker Images

Topik: Build images in CI, Docker Hub, image tagging

Proyek: DockerCIPipeline

Build Docker images in CI

- Build image in GitHub Actions
- Tag dengan version & latest
- Push to Docker Hub (simulate)
- Multi-platform builds
- Cache Docker layers
- Teknologi: Docker CI/CD

Hari 20: Deployment Automation

Topik: Deploy to server from CI, SSH deployment

Proyek: AutomatedDeployment

Automated deployment pipeline

- SSH into server from CI (simulate)
- Pull latest code
- Run migrations
- collectstatic
- Restart Gunicorn
- Rollback on failure
- Teknologi: Deployment automation

Hari 21: Review & Mini Project

Mini Project: FullCICDPipeline

Complete CI/CD untuk Django project

Pipeline stages:

1. Lint (flake8, black)
2. Test (pytest, coverage > 80%)
3. Build Docker image
4. Push to registry
5. Deploy to staging (simulate)
6. Integration tests
7. Deploy to production (manual approval simulate)

Features:

- .github/workflows/main.yml
- Separate workflows (CI, CD)
- Environment secrets
- Deployment notifications (simulate)
- Rollback mechanism
- Database backup before deploy
- Blue-green deployment simulation
- Teknologi: GitHub Actions, Docker, Automated deployment

Minggu 4: Cloud Deployment & Monitoring

Hari 22: Cloud Platforms Overview

Topik: AWS, GCP, Azure, Heroku, DigitalOcean comparison

Proyek: CloudPlatformComparison

Evaluate cloud platforms

- Research different platforms
- Pricing comparison
- Features comparison
- Django hosting options
- Choose platform for deployment (DigitalOcean simulate)
- Teknologi: Cloud platforms research

Hari 23: Server Setup & Provisioning

Topik: VPS setup, SSH, firewall, basic server security

Proyek: ServerProvisioning

Setup production server

- VPS setup (simulate Ubuntu server)
- SSH key authentication
- UFW firewall configuration
- Fail2ban setup (simulate)
- User management
- Basic server hardening
- Teknologi: Linux server administration

Hari 24: Deployment to VPS

Topik: Deploy Django to VPS, systemd services

Proyek: VPSDeployment

Deploy to virtual private server

- Clone repository
- Setup virtual environment
- Install dependencies
- Configure Gunicorn systemd service
- Configure Nginx
- Setup PostgreSQL
- Run migrations & collectstatic
- Teknologi: VPS deployment

Hari 25: Application Monitoring

Topik: Health checks, uptime monitoring, alerting

Proyek: AppMonitoring

Monitor application health

- Health check endpoint
- Uptime monitoring (UptimeRobot simulate)
- Status page

- Alert on downtime (email simulate)
- Response time monitoring
- Teknologi: Application monitoring

Hari 26: Error Tracking & Logging

Topik: Sentry integration, error aggregation, debugging production

Proyek: ErrorTracking

Production error tracking

- Sentry setup (simulate)
- Capture exceptions
- Error grouping
- Stack traces
- User context
- Release tracking
- Teknologi: Sentry, error tracking

Hari 27: Performance Monitoring

Topik: APM tools, New Relic, performance metrics

Proyek: PerformanceMonitoring

Monitor application performance

- New Relic (simulate) setup
- Track response times
- Database query performance
- Slow endpoint detection
- Custom metrics
- Performance alerts
- Teknologi: APM, performance monitoring

Hari 28: Review & Final Project Bulan 6

Final Project: EnterpriseDeployment

Enterprise-grade deployment setup

Application: E-commerce API (from previous months)

Infrastructure:

- Multi-container Docker setup
- PostgreSQL database (separate container)
- Redis for caching & Celery
- Celery workers
- Nginx reverse proxy
- SSL/TLS setup

CI/CD:

- GitHub Actions pipeline
- Automated testing (>80% coverage)
- Code quality checks
- Docker image builds
- Automated deployment

Deployment:

- VPS deployment (simulate DigitalOcean)
- Gunicorn + Nginx
- PostgreSQL production database
- Systemd services
- Environment-based configuration

Monitoring:

- Health check endpoints
- Sentry error tracking (simulate)
- Application monitoring (simulate)
- Log aggregation
- Uptime monitoring
- Performance monitoring

Security:

- SSL certificate
- Security headers

- Rate limiting
- Database backups automated
- Secrets management

Documentation:

- Deployment guide
- Architecture diagram
- API documentation
- Runbook for common issues

Teknologi: Docker, CI/CD, VPS, Nginx, PostgreSQL, Monitoring, Security

Progress Check Bulan 6: ✓ Production settings & configuration ✓ Static files in production ✓ Gunicorn & WSGI ✓ Nginx reverse proxy ✓ SSL/TLS & HTTPS ✓ Docker & containerization ✓ Docker Compose multi-container ✓ CI/CD dengan GitHub Actions ✓ Automated testing & linting ✓ Deployment automation ✓ VPS deployment ✓ Application monitoring ✓ Error tracking & logging ✓ Performance monitoring

Total Proyek: 60+ proyek & 4 major deployments **Siap lanjut:** Bulan 7 - Advanced Patterns & Scalability